Plot plan, elevations and piping isometrics

Flash drum D-400 unit · plot plan with spacing rules, D-400 elevation and levels, bottoms static head, five isometric sheets with bills of materials, operator round

Document no. FD-V1-LA-PLT-001

Sheet 1 places every item of the equipment list on the shared plot plan (positions of core/plot.json kept) and checks the spacing rules (typical practice, assumed). Sheets 2–4 give the D-400 elevations, platforms, ladders, levels and reading heights, the static head of the bottoms line and the drain fall. Sheets 5–9 are the isometrics of the main lines with their bills of materials; sheet 10 the outside-operator round and the hand-valve positions that the 3-D model (layout.json) uses.

Contents

Sheet 1 · Plot plan and locations
Sheet 2 · Spacing and layout rules
Sheet 3 · D-400 elevation and levels
Sheet 4 · Bottoms static head, drain, reading levels
Sheets 5–9 · Piping isometrics and bills of materials
Sheet 10 · Operator round, hand valves, layout.json
Sheet 11 · Checks and engineering queries

Holds and engineering queries

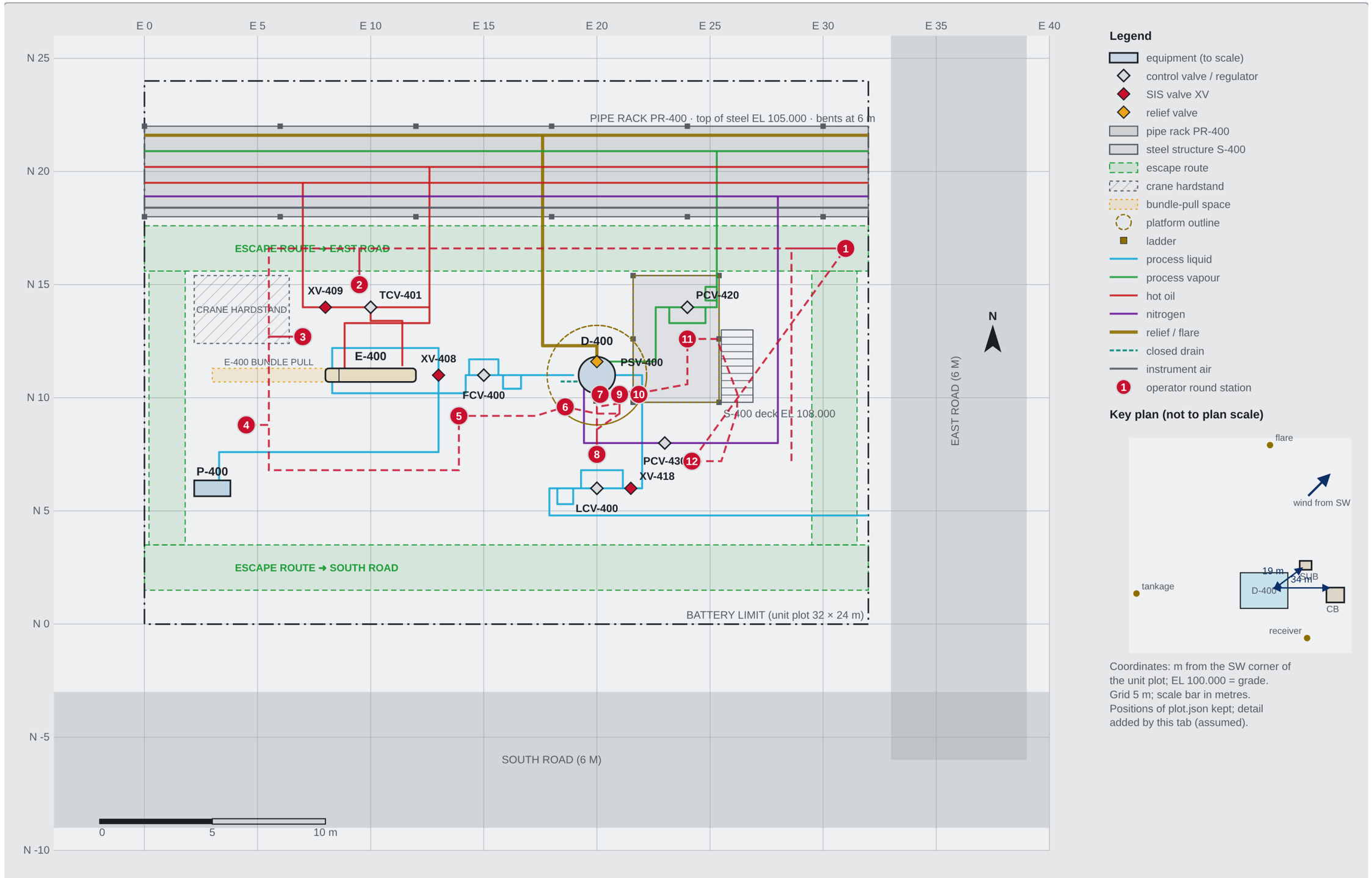
- LA-01 XV-408 position on the plot plan
- LA-02 Static head datum of the model (K.Hel)
- LA-03 Two-phase riser after FCV-400
- LA-04 LSLL-417 not reachable for proof testing
- LA-05 Siting of the control building and the substation
- LA-06 Bottoms line flashes at FT-410 at high flow and low level
- LA-07 Field items without a tag number

Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

Rev	Date	Description	Originator	Checked	Approved
0	28-OCT-26	Issued for teaching	A. Soottitantawat	—	—

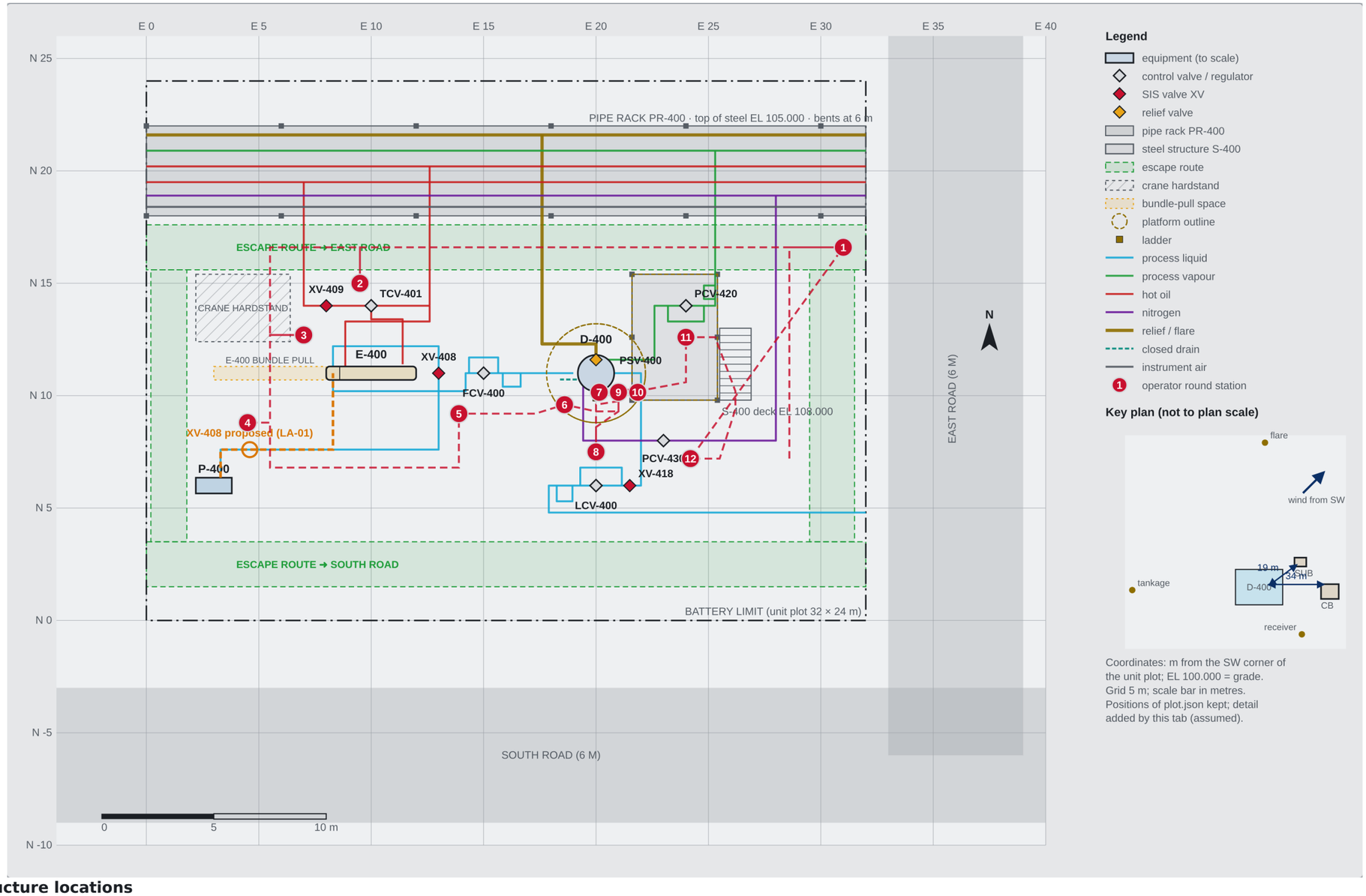
DOCUMENT NO.	REV	STATUS	ORIGINATOR	DATE
FD-V1-LA-PLT-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

Sheet 1 · Plot plan (positions of core/plot.json kept; detail added, assumed)



Coordinates: m from the SW corner of the unit plot; EL 100.000 = grade. Grid 5 m; scale bar in metres. Positions of plot.json kept; detail added by this tab (assumed).

Sheet 1 (cont.) · Query LA-01: XV-408 where FD-400-PID-001 puts it (upstream of E-400, dashed)



Equipment, valve and structure locations

Tag	Item	Shape	x (E) m	y (N) m	Bottom / CL EL	Top EL	Footprint m	Source	Basis
D-400	Flash drum	vertical cylinder	20.00	11.00	EL 100.000	EL 108.200	1.62 × 1.62	plot.json	1600 mm ID × 4800 mm T/T, shell 8 mm (EQL); bottom tangent EL 103.000 (plot.json), skirt to grade; uninsulated
E-400	Feed preheater (hot oil)	horizontal cylinder (x-axis)	10.00	11.00	EL 100.900	EL 101.500	4.00 × 0.60	plot.json	TEMA BEU, shell 305 mm ID × 4 m (EQL); plot.json z 1.2 m read as the centreline (assumed); channel and bundle pull to the west
P-400	Feed supply pump (upstream unit, interface)	box	3.00	6.00	EL 100.000	EL 100.900	1.60 × 0.70	plot.json	upstream unit scope, shown at the battery limit (plot.json); baseplate 1.6 × 0.7 m, height 0.9 m (assumed)
CB	Control building (operators)	box	64.00	9.00	EL 100.000	EL 104.500	12.00 × 10.00	plot.json	plot.json
SUB	Electrical substation	box	44.00	29.00	EL 100.000	EL 104.000	8.00 × 6.00	plot.json	plot.json
FCV-400	Feed let-down valve	valve (tag)	15.00	11.00	EL 101.000 (CL)	—	—	plot.json	2"-P-4002-A1A · Globe (angle body for flashing service), equal-% trim, R = 30, fail closed
TCV-401	Hot-oil 3-way valve	valve (tag)	10.00	14.00	EL 101.000 (CL)	—	—	plot.json	2"-HO-4011-B1A · 3-way diverting, linear, fail to full bypass
LCV-400	Bottoms level valve	valve (tag)	20.00	6.00	EL 100.800 (CL)	—	—	plot.json	2"-P-4003-A1A · Globe, equal-% trim, fail closed (flashing outlet)
PCV-420	Vapour pressure-control valve	valve (tag)	24.00	14.00	EL 109.000 (CL)	—	—	plot.json	4"-PV-4021-A1A·a · Globe (or butterfly), equal-%, fail open
PCV-430	Nitrogen blanket regulator	valve (tag)	23.00	8.00	EL 101.000 (CL)	—	—	plot.json	1"-N-4031-A1A·a · Self-acting pressure regulator, set 30 kPa g
PSV-400	D-400 relief valve	valve (tag)	20.00	11.60	EL 109.200 (CL)	—	—	plot.json	3"-PV-4023-A1A · Conventional spring-loaded, closed bonnet, to flare
XV-408	Feed SIS shut-off valve	valve (tag)	13.00	11.00	EL 101.000 (CL)	—	—	plot.json	2"-P-4001-A1A · Ball, full bore, spring-return actuator, fail closed, tight shut-off (class V)
XV-409	Hot-oil SIS shut-off valve	valve (tag)	8.00	14.00	EL 101.000 (CL)	—	—	plot.json	2"-HO-4011-B1A · Ball, fail closed, fire-safe
XV-418	Bottoms SIS shut-off valve	valve (tag)	21.50	6.00	EL 100.800 (CL)	—	—	plot.json	2"-P-4003-A1A · Ball, fail closed, tight shut-off (gas blow-by)
PR-400	Pipe rack	structure	16.00	20.00	EL 100.000	EL 105.000	32.0 × 4.0	plot.json	single tier, top of steel EL 105.000 (plot.json); bents every 6 m (assumed)
S-400	Valve-station structure	structure	23.50	12.60	EL 100.000	EL 108.000	3.8 × 5.6	added	steel structure added by the layout (assumed) to carry the PCV-420 station at plot.json EL 109.000 next to the D-400 top platform

Tag	Item	Shape	x (E) m	y (N) m	Bottom / CL EL	Top EL	Footprint m	Source	Basis
PL-1	platform on D-400	ring 0° to 360°	20.00	11.00	EL 105.600	—	r 0.9–2.2	added	feed inlet N1, manway M1, LSHH-416 (N7), LG-400 upper half
PL-2	platform on D-400	ring -150° to 150°	20.00	11.00	EL 108.000	—	r 0.9–2.2	added	top head: N2, PSV-400, PT-420, PSHH-426, PG-420, AI-400, N5 (open to the west for the PSV tail pipe)
S-400-1	platform on S-400	deck	23.50	12.60	EL 108.000	—	3.8 × 5.6	added	PCV-420 station, HV-421/422/423, FI-420, AI-420

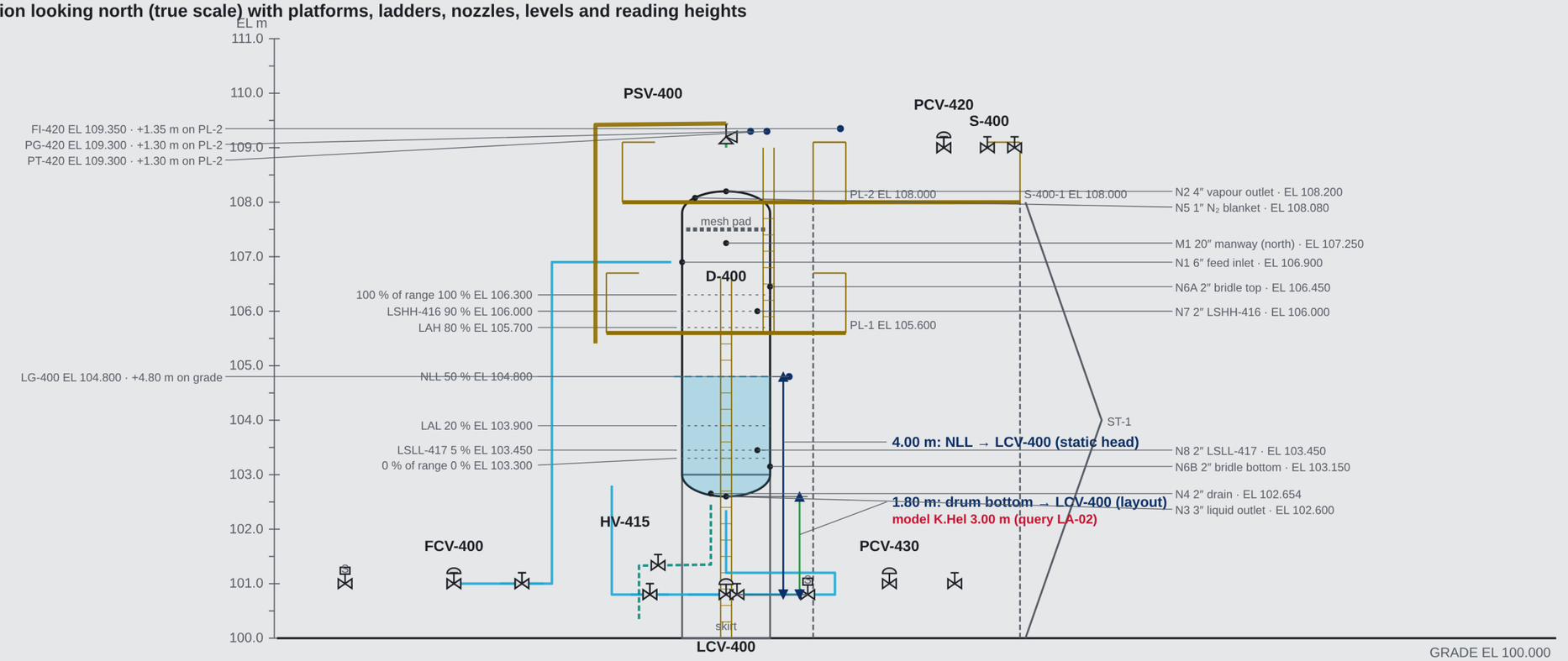
Sheet 2 · Spacing and layout rules (typical practice, assumed)

Rule	Typical practice (assumed)	Limit	Pair / item	Value	Result
SR-01	Vessel to exchanger, shell to shell (plan)	3 m	D-400 / E-400	7.19 m	✓
SR-02	Pump handling flammables to vessels and exchangers	3 m	P-400 / E-400	6.05 m	✓
			P-400 / D-400	16.05 m	✓
SR-03	Exchanger tube-bundle pull space: tube length + 1.0 m, clear of equipment and pipes below 2 m	1 m beyond the tubes	E-400 west: clear 8.0 m to battery limit; pull space 5.0 m (plot.json)	5.0 m vs 4.9 m	✓
SR-04	Crane hardstand within this distance of the pulled bundle, and clear	8 m (max)	HS-1 centre to the bundle centre 3.1 m; free of equipment and pipes	3.1 m	✓
SR-05	Unit equipment to the battery limit (P-400 is upstream scope)	3 m	nearest: E-400	8.00 m	✓
SR-06	Hydrocarbon equipment to an occupied, non-blast-resistant building	60 m	PCV-420 → control building CB	33.6 m	✗
SR-07	Hydrocarbon equipment to the substation (ignition source)	30 m	PCV-420 → substation SUB	19.4 m	✗
SR-08	Occupied building upwind: its bearing within ± this angle of the wind-from direction	45 ° (max)	bearing D-400 → CB 93°, wind from 225°	132° apart	✗
SR-09	Ignition source not downwind: bearing at least this far from the downwind direction	45 °	bearing D-400 → SUB 53°, downwind 45°	8° from downwind	✗
SR-10	Escape route clear width	1.2 m	ES-N North aisle (2 m) to the east road	2.0 m	✓
			ES-S South aisle (2 m) to the south road	2.0 m	✓
			ES-E East aisle (2 m)	2.0 m	✓
			ES-W West walkway (1.6 m)	1.6 m	✓
SR-11	Headroom under pipes crossing escape routes	2.2 m	6 crossings; lowest 4"-P-4005-A1A over ES-E	2.74 m	✓
SR-12	Headroom over the maintenance access way under the rack	4 m	AW-1: 10 crossings; lowest 2"-HO-4011-B1A; rack steel EL 105.000	4.27 m	✓
SR-13	Travel distance (plan) from any equipment item or round station to an escape route	15 m (max)	farthest: station 6	6.0 m	✓
SR-14	Escape routes from each elevated floor (stair or ladder)	2 routes	top floor EL 108.000 (PL-2 + S-400-1): stair ST-1 + ladders LD-2/LD-1	2 routes	✓
			PL-1 EL 105.600: LD-1 down, LD-2 up to the stair	2 routes	✓
SR-15	Ladder flight between landings	6 m (max)	longest flight: LD-1 (EL 100.000 → EL 105.600)	5.60 m	✓
SR-16	Relief tail pipe falls to the flare header (no pocket), slope at least	0.005 m/m	6"-FL-4041-F1A: 0 rise(s) along the flow; smallest fall 0.0091 m/m (1 : 110)	0.0091	✓
SR-17	Gravity drain falls to the closed drain (no pocket), slope at least	0.01 m/m	2"-CD-4051-A1A: 0 rise(s) along the flow; smallest fall 0.0152 m/m (1 : 66)	0.0152	✓
SR-18	Let-down valve close to the drum: two-phase line FCV-400 → N1 at most	10 m (max)	6"-P-4004-A1A route length (rises 5.9 m, query LA-03)	9.89 m	✓
SR-19	Hot-oil supply below the auto-ignition temperature of the process liquid by at least	25 K	hot oil 180 °C vs n-hexane AIT 225 °C (typical)	45 K	✓
SR-20	Elevated flare to process equipment	60 m	flare (plot.json) to D-400	98 m	✓

21 ✓, 4 ✗. Change a limit to see which pairs would fail; Show draws the distance on the plot plan. Every ✗ comes from a position of the shared plot plan that this tab may not move: it is a query (LA-05), not a correction.

Sheet 3 · D-400 elevation looking north

D-400 · elevation looking north (true scale) with platforms, ladders, nozzles, levels and reading heights



D-400 levels and elevations

Level / item	% of LT-400 range	EL	above bottom tangent	above grade
Drum bottom (head apex)	—	EL 102.600	−0.40 m	2.60 m
Bottom tangent	—	EL 103.000	0.00 m	3.00 m
0 % of range	0 %	EL 103.300	0.30 m	3.30 m
LSLL-417	5 %	EL 103.450	0.45 m	3.45 m
LAL	20 %	EL 103.900	0.90 m	3.90 m
NLL	50 %	EL 104.800	1.80 m	4.80 m
LAH	80 %	EL 105.700	2.70 m	5.70 m
LSHH-416	90 %	EL 106.000	3.00 m	6.00 m
100 % of range	100 %	EL 106.300	3.30 m	6.30 m
Feed inlet N1	—	EL 106.900	3.90 m	6.90 m
Mesh pad	—	EL 107.500	4.50 m	7.50 m
Top tangent	—	EL 107.800	4.80 m	7.80 m
PL-1 / PL-2 / S-400 deck	—	EL 105.600 / EL 108.000		
LCV-400 / XV-418 centreline	—	EL 100.800		0.80 m

Sheet 4 · Bottoms static head, model datum, drain and reading levels

Static head to LCV-400 and the model datum (query LA-02, answers EM-01). The model computes the bottoms driving pressure as Pdrum + ρg(K.Hel + hliq) − Preceiver − line loss, with hliq measured from the drum bottom (hliq = 2.20 m at 50 %). K.Hel is therefore the height of the drum bottom above the LCV-400 inlet. The layout gives EL 102.600 − EL 100.800 = 1.80 m; the model uses 3.00 m.

The 1.2 m difference costs 7.1 kPa of head and about 1.3 % of valve opening at design: the model is slightly optimistic, the layout is kept (plot.json) and the model constant is to be revised (LA-02).

	Model (K.Hel 3.00 m)	Layout (1.80 m)
Static head at NLL, ρ 605.4 kg/m³	30.88 kPa	23.76 kPa
LCV-400 ΔP at design (Pd 102.7 + head − 40 − 10 kPa line)	83.6 kPa	76.5 kPa
LCV-400 opening for the design flow (equal-%, R 30, √ΔP)	63.4 % (model)	64.7 %

Bottoms line at design (2″ as drawn, NLL)

Quantity	Value	How
Velocity in 2″ Sch 40	1.80 m/s	Q / (π D²/4)
Reynolds number / Churchill f	291,314 / 0.0202	ρ v D / μ
Static head, liquid surface → FT-410 (EL 101.200)	21.38 kPa	ρ g (zliq − zFT)
Entrance and acceleration (1 + Kin) ρv²/2	1.47 kPa	Kin 0.5 (vortex breaker + nozzle, typical)
Friction N3 → FT-410	2.58 kPa	f (L + 2 × 20 D)/D · ρv²/2
FT-410 differential (vena contracta)	7.84 kPa	25 kPa at 25 m³/h full scale (assumed) × (Q/FS)²
Margin above the bubble pressure at the vena contracta	9.49 kPa ✓	head − entrance − friction − orifice ΔP
Margin at the LCV-400 inlet (EL 100.800)	11.84 kPa ✓	head − entrance − friction − XV-418 / block − orifice permanent loss (0.6 ΔP)
Lowest drum level that keeps FT-410 liquid	EL 103.201 = −3 % of the LT-400 range	zFT + (losses)/(ρg)

Margin at the FT-410 vena contracta, kPa (2″ / 4″)

Drum level	EL of liquid	14.0 m³/h (design) 2″ / 4″ margin kPa	17.5 m³/h (maximum rate) 2″ / 4″ margin kPa	18.0 m³/h (FIC-410 HI alarm) 2″ / 4″ margin kPa
LSLL-417 5 %	EL 103.450	1.4 ✓ / 5.3 ✓	−5.2 ✗ / 0.7 ✓	−6.2 ✗ / 0.1 ✓
LAL 20 %	EL 103.900	4.1 ✓ / 7.9 ✓	−2.6 ✗ / 3.4 ✓	−3.6 ✗ / 2.7 ✓
NLL 50 %	EL 104.800	9.4 ✓ / 13.3 ✓	2.8 ✓ / 8.8 ✓	1.8 ✓ / 8.1 ✓
LAH 80 %	EL 105.700	14.8 ✓ / 18.6 ✓	8.1 ✓ / 14.1 ✓	7.1 ✓ / 13.4 ✓

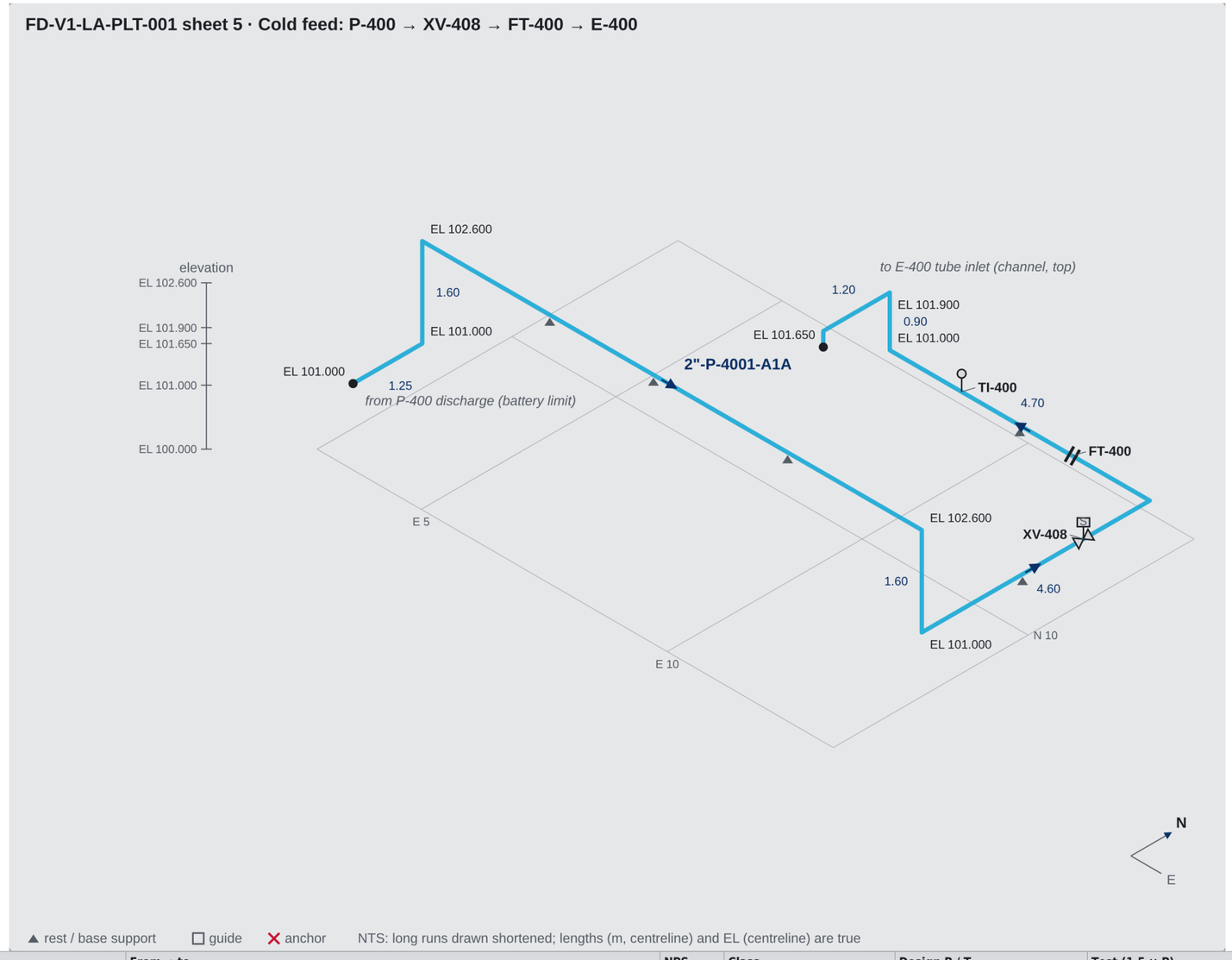
Gravity drain to HV-415. The drain N4 (2″, bottom head, EL 102.450) drops inside the skirt to EL 101.350, leaves the skirt to the west falling 0.020 m over 1.32 m (1 : 66, rule ≥ 1 : 100, typical) through HV-415 at EL 101.335 — within reach from grade — and drops into the closed-drain hub at EL 100.300. No pocket (0 rises), so the line empties when HV-415 is opened and the drum can be drained to the last litre.

Instrument reading levels and hand-valve heights

Tag	Device	Kind	Where	EL	Floor	Height above floor	Reading / operating	
XV-408		SIS valve	2"-P-4001-A1A	EL 101.000	grade	1.00 m	access for maintenance	
FT-400		transmitter	2"-P-4001-A1A	EL 101.350	grade	1.35 m	read (m³/h)	✓
TI-400	TT-400	thermowell + transmitter	2"-P-4001-A1A	EL 101.250	grade	1.25 m	read (°C)	✓
TT-401		thermowell + transmitter	2"-P-4002-A1A	EL 100.550	grade	0.55 m	read (°C)	✓
BV-FCV400-U not on P&ID		station block valve	2"-P-4002-A1A	EL 101.000	grade	1.00 m	maintain / test	✓
FCV-400		control valve	2"-P-4002-A1A	EL 101.000	grade	1.00 m	access for maintenance	
HV-403		hand valve	2"-P-4002-A1A	EL 101.000	grade	1.00 m	operate (NC, throttling)	✓
BV-FCV400-D not on P&ID		station block valve	6"-P-4004-A1A	EL 101.000	grade	1.00 m	maintain / test	✓
HV-401		hand valve	6"-P-4004-A1A	EL 101.000	grade	1.00 m	operate (NO)	✓
HV-402		hand valve	6"-P-4004-A1A	EL 101.000	grade	1.00 m	operate (NC)	✓
PG-420 not on P&ID		local gauge	4"-PV-4021-A1A·a	EL 109.300	PL-2 EL 108.000	1.30 m	read (kPa g)	✓
PT-420		transmitter	4"-PV-4021-A1A·a	EL 109.300	PL-2 EL 108.000	1.30 m	read (kPa g)	✓
PSHH-426		transmitter	4"-PV-4021-A1A·a	EL 109.300	PL-2 EL 108.000	1.30 m	maintain / test	✓
AI-400		sample point	4"-PV-4021-A1A·a	EL 109.300	PL-2 EL 108.000	1.30 m	maintain / test	✓
FI-420	FT-420	transmitter	4"-PV-4021-A1A·a	EL 109.350	PL-2 EL 108.000	1.35 m	read (m³/h)	✓
AI-420		sample point	4"-PV-4021-A1A·a	EL 109.300	S-400-1 EL 108.000	1.30 m	maintain / test	✓
BV-PCV420-U not on P&ID		station block valve	4"-PV-4021-A1A·a	EL 109.000	S-400-1 EL 108.000	1.00 m	maintain / test	✓
PCV-420		control valve	4"-PV-4021-A1A·a	EL 109.000	S-400-1 EL 108.000	1.00 m	access for maintenance	
HV-423		hand valve	4"-PV-4021-A1A·a	EL 109.000	S-400-1 EL 108.000	1.00 m	operate (NC, throttling)	✓
BV-PCV420-D not on P&ID		station block valve	4"-PV-4021-A1A·b	EL 109.000	S-400-1 EL 108.000	1.00 m	maintain / test	✓
HV-421		hand valve	4"-PV-4021-A1A·b	EL 109.000	S-400-1 EL 108.000	1.00 m	operate (NO)	✓
HV-422		hand valve	4"-PV-4021-A1A·b	EL 109.000	S-400-1 EL 108.000	1.00 m	operate (NC)	✓
PSV-400		relief valve	3"-PV-4023-A1A	EL 109.200	PL-2 EL 108.000	1.20 m	access for maintenance	
FT-410		transmitter	2"-P-4003-A1A	EL 101.550	grade	1.55 m	read (m³/h)	✓
AI-410		sample point	2"-P-4003-A1A	EL 101.500	grade	1.50 m	maintain / test	✓
TI-410	TT-410	thermowell + transmitter	2"-P-4003-A1A	EL 101.450	grade	1.45 m	read (°C)	✓
XV-418		SIS valve	2"-P-4003-A1A	EL 100.800	grade	0.80 m	access for maintenance	
BV-LCV400-U not on P&ID		station block valve	2"-P-4003-A1A	EL 100.800	grade	0.80 m	maintain / test	✓
LCV-400		control valve	2"-P-4003-A1A	EL 100.800	grade	0.80 m	access for maintenance	
HV-413		hand valve	2"-P-4003-A1A	EL 100.800	grade	0.80 m	operate (NC, throttling)	✓
BV-LCV400-D not on P&ID		station block valve	4"-P-4005-A1A	EL 100.800	grade	0.80 m	maintain / test	✓
HV-411		hand valve	4"-P-4005-A1A	EL 100.800	grade	0.80 m	operate (NO)	✓
HV-412		hand valve	4"-P-4005-A1A	EL 100.800	grade	0.80 m	operate (NC)	✓
HV-415		hand valve	2"-CD-4051-A1A	EL 101.335	grade	1.34 m	operate (NC)	✓
HV-430		hand valve	1"-N-4031-A1A·a	EL 101.000	grade	1.00 m	operate (NO)	✓
PCV-430		self-acting regulator	1"-N-4031-A1A·a	EL 101.000	grade	1.00 m	access for maintenance	
XV-409		SIS valve	2"-HO-4011-B1A	EL 101.000	grade	1.00 m	access for maintenance	
TI-402	TT-402	thermowell + transmitter	2"-HO-4011-B1A	EL 101.250	grade	1.25 m	read (°C)	✓
PDI-406		transmitter	2"-HO-4011-B1A	EL 101.300	grade	1.30 m	read (kPa)	✓
PG-406 not on P&ID		local gauge	2"-HO-4011-B1A	EL 101.300	grade	1.30 m	read (kPa g)	✓
TCV-401		control valve	2"-HO-4011-B1A	EL 101.000	grade	1.00 m	access for maintenance	
FT-405		transmitter	3"-HO-4013-B1A	EL 100.950	grade	0.95 m	read (m³/h)	✓
TI-403	TT-403	thermowell + transmitter	3"-HO-4015-B1A	EL 100.530	grade	0.53 m	read (°C)	✓
LG-400		local gauge	D-400	EL 104.800	grade	4.80 m	read (%)	✓
LT-400		transmitter	D-400	EL 101.400	grade	1.40 m	read (%)	✓
LSHH-416		transmitter	D-400	EL 106.000	PL-1 EL 105.600	0.40 m	maintain / test	✓

Tag	Device	Kind	Where	EL	Floor	Height above floor	Reading / operating	
LSLL-417		transmitter	D-400	EL 103.450	grade	3.45 m	maintain / test	✗

Sheet 5 · Cold feed: P-400 → XV-408 → FT-400 → E-400



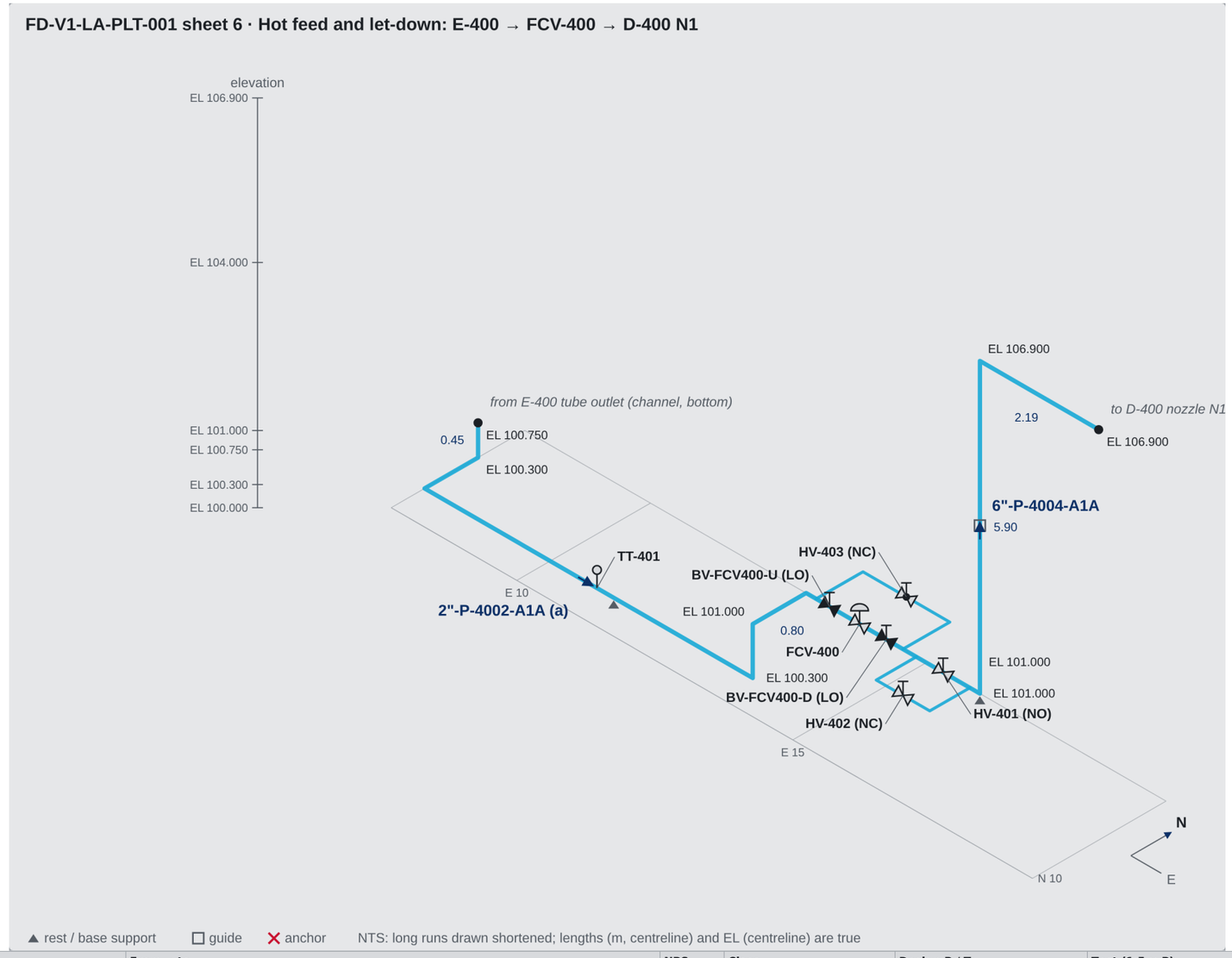
Line	Service	From → to	NPS	Class	Design P / T	Test (1.5 × P)	Insulation	Route length	Line list
2"-P-4001-A1A	Cold feed	P-400 discharge (battery limit) → E-400 tube inlet (channel, top)	2″	A1A · class 150	820 kPa g / 65 °C	1,230 kPa g	AC (anti-condensation 25 mm)	25.8 m	✓

Worked example (2"-P-4001-A1A). Route length = sum of the centreline segments of the main run: 1.25 + 1.60 + 9.70 + 1.60 + 4.60 + 4.70 + 0.90 + 1.20 + 0.25 = 25.80 m; ordered = route × 1.05 (cut and weld allowance, assumed) = 27.1 m. Supports: rests at the suggested span for 2″ (3 m, typical) on horizontal runs, guides every 6 m on risers, the rack bents on the rack: 5 supports. Hydrotest (ASME B31.3, 1.5 × design, temperature correction ignored) = 1.5 × 820 = 1,230 kPa g.

Bill of materials · 2"-P-4001-A1A · 2″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	2″	27.1	m	route length 25.8 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	2″	8	ea	
Flange WN RF, class 150	2″	4	ea	equipment nozzles 2 + 2 per flanged valve
Orifice flange pair WN RF class 150 + plate	2″	1	set	FT-400
Gasket spiral-wound, class 150	2″	6	ea	
Stud bolt set, class 150	2″	6	set	
Ball valve, class 150, spring-return actuator (by instruments)	2″	1	ea	XV-408
Thermowell 1½″ flanged	1½″	1	ea	TI-400
Support: Rest (shoe)	2″	5	ea	
Insulation AC (anti-condensation 25 mm)	2″	26.0	m	

Sheet 6 · Hot feed and let-down: E-400 → FCV-400 → D-400 N1



Line	Service	From → to	NPS	Class	Design P / T	Test (1.5 × P)	Insulation	Route length	Line list
2"-P-4002-A1A (a)	Hot feed, liquid under pressure	E-400 tube outlet (channel, bottom) → FCV-400	2″	A1A · class 150	820 kPa g / 205 °C	1,230 kPa g	H (heat conservation, mineral wool 50 mm)	12.2 m	✓
6"-P-4004-A1A	Let-down outlet, flashing two-phase (drawn on the P&ID as part of 2"-P-4002-A1A)	FCV-400 outlet → D-400 nozzle N1 (6″, EL 106.900)	6″	A1A · class 150	450 kPa g / 150 °C	675 kPa g	— (none)	11.9 m	✓ proposed

Worked example (2"-P-4002-A1A). Route length = sum of the centreline segments of the main run: 0.45 + 0.80 + 5.90 + 0.70 + 0.80 + 0.80 = 9.45 m, plus 1 branch run(s) 2.70 m; ordered = route × 1.05 (cut and weld allowance, assumed) = 12.8 m. Supports: rests at the suggested span for 2″ (3 m, typical) on horizontal runs, guides every 6 m on risers, the rack bents on the rack: 1 supports. Hydrotest (ASME B31.3, 1.5 × design, temperature correction ignored) = 1.5 × 820 = 1,230 kPa g.

Bill of materials · 2"-P-4002-A1A (a) · 2″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

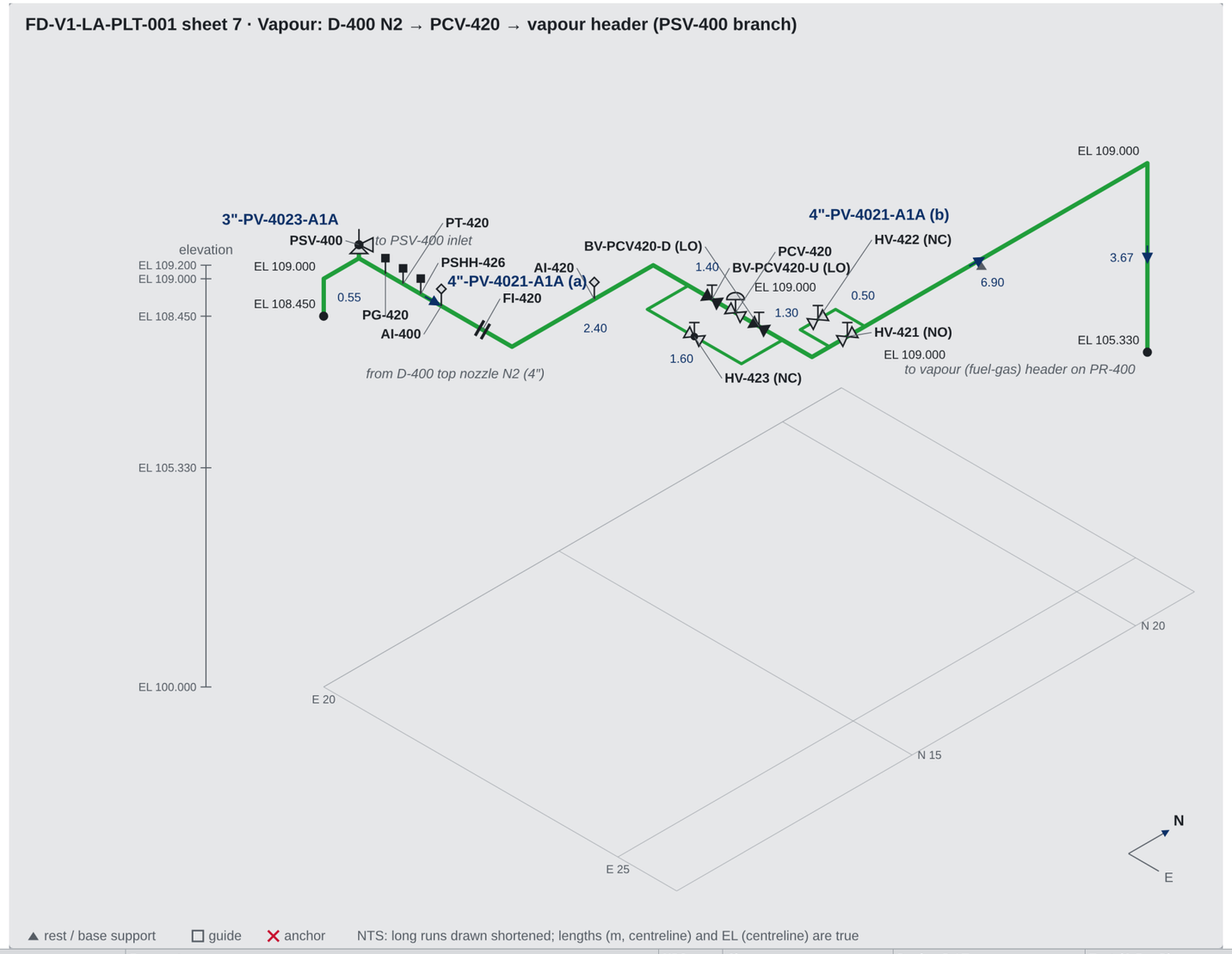
Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	2″	12.8	m	route length 12.2 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	2″	7	ea	
Tee, equal, BW	2″	2	ea	branch runs (bypass loops / branches)
Flange WN RF, class 150	2″	7	ea	equipment nozzles 1 + 2 per flanged valve
Gasket spiral-wound, class 150	2″	7	ea	
Stud bolt set, class 150	2″	7	set	
Gate valve, class 150 (station block, locked open)	2″	1	ea	BV-FCV400-U
Globe valve, class 150 (throttling bypass)	2″	1	ea	HV-403
Control valve, class 150 (by instruments)	2″	1	ea	FCV-400
Thermowell 1½″ flanged	1½″	1	ea	TT-401
Support: Rest (shoe)	2″	1	ea	

Item	Size	Qty	Unit	Note
Insulation H (heat conservation, mineral wool 50 mm)	2″	13.0	m	

Bill of materials · 6"-P-4004-A1A · 6″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	6″	12.5	m	route length 11.9 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	6″	4	ea	
Tee, equal, BW	6″	2	ea	branch runs (bypass loops / branches)
Flange WN RF, class 150	6″	7	ea	equipment nozzles 1 + 2 per flanged valve
Gasket spiral-wound, class 150	6″	7	ea	
Stud bolt set, class 150	6″	7	set	
Gate valve, class 150 (station block, locked open)	6″	1	ea	BV-FCV400-D
Gate valve, class 150 (hand valve)	6″	2	ea	HV-401, HV-402
Support: Guide	6″	1	ea	
Support: Trunnion / base support	6″	1	ea	

Sheet 7 · Vapour: D-400 N2 → PCV-420 → vapour header (PSV-400 branch)



Line	Service	From → to	NPS	Class	Design P / T	Test (1.5 × P)	Insulation	Route length	Line list
4"-PV-4021-A1A (a)	Vapour, drum to the pressure-control valve (PSV-400 branch)	D-400 top nozzle N2 (4″) → PCV-420	4″	A1A · class 150	450 kPa g / 150 °C	675 kPa g	— (none)	10.6 m	✓
4"-PV-4021-A1A (b)	Vapour after PCV-420	PCV-420 outlet → vapour (fuel-gas) header on PR-400	4″	A1A · class 150	450 kPa g / 150 °C	675 kPa g	— (none)	13.5 m	✓
3"-PV-4023-A1A	PSV-400 inlet (not numbered on the P&ID)	tee on 4″-PV-4021-A1A → PSV-400 inlet	3″	A1A · class 150	450 kPa g / 150 °C	675 kPa g	— (none)	0.2 m	✓ proposed

Worked example (4"-PV-4021-A1A). Route length = sum of the centreline segments of the main run: 0.55 + 0.60 + 2.60 + 2.40 + 1.40 = 7.55 m, plus 1 branch run(s) 3.00 m; ordered = route × 1.05 (cut and weld allowance, assumed) = 11.1 m. Supports: rests at the suggested span for 4″ (4.3 m, typical) on horizontal runs, guides every 6 m on risers, the rack bents on the rack: 0 supports. Hydrotest (ASME B31.3, 1.5 × design, temperature correction ignored) = 1.5 × 450 = 675 kPa g.

Bill of materials · 4"-PV-4021-A1A (a) · 4″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	4″	11.1	m	route length 10.6 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	4″	6	ea	
Tee, reducing, BW	4″ × 2″	2	ea	
Flange WN RF, class 150	4″	7	ea	equipment nozzles 1 + 2 per flanged valve
Orifice flange pair WN RF class 150 + plate	4″	1	set	FI-420
Gasket spiral-wound, class 150	4″	9	ea	
Stud bolt set, class 150	4″	9	set	
Gate valve, class 150 (station block, locked open)	4″	1	ea	BV-PCV420-U
Globe valve, class 150 (throttling bypass)	4″	1	ea	HV-423
Control valve, class 150 (by instruments)	4″	1	ea	PCV-420
Pressure tap: ¾″ weldolet + root valve	¾″	3	ea	PG-420, PT-420, PSHH-426

Item	Size	Qty	Unit	Note
Sample connection: ¾″ weldolet + root valve	¾″	2	ea	AI-400, AI-420

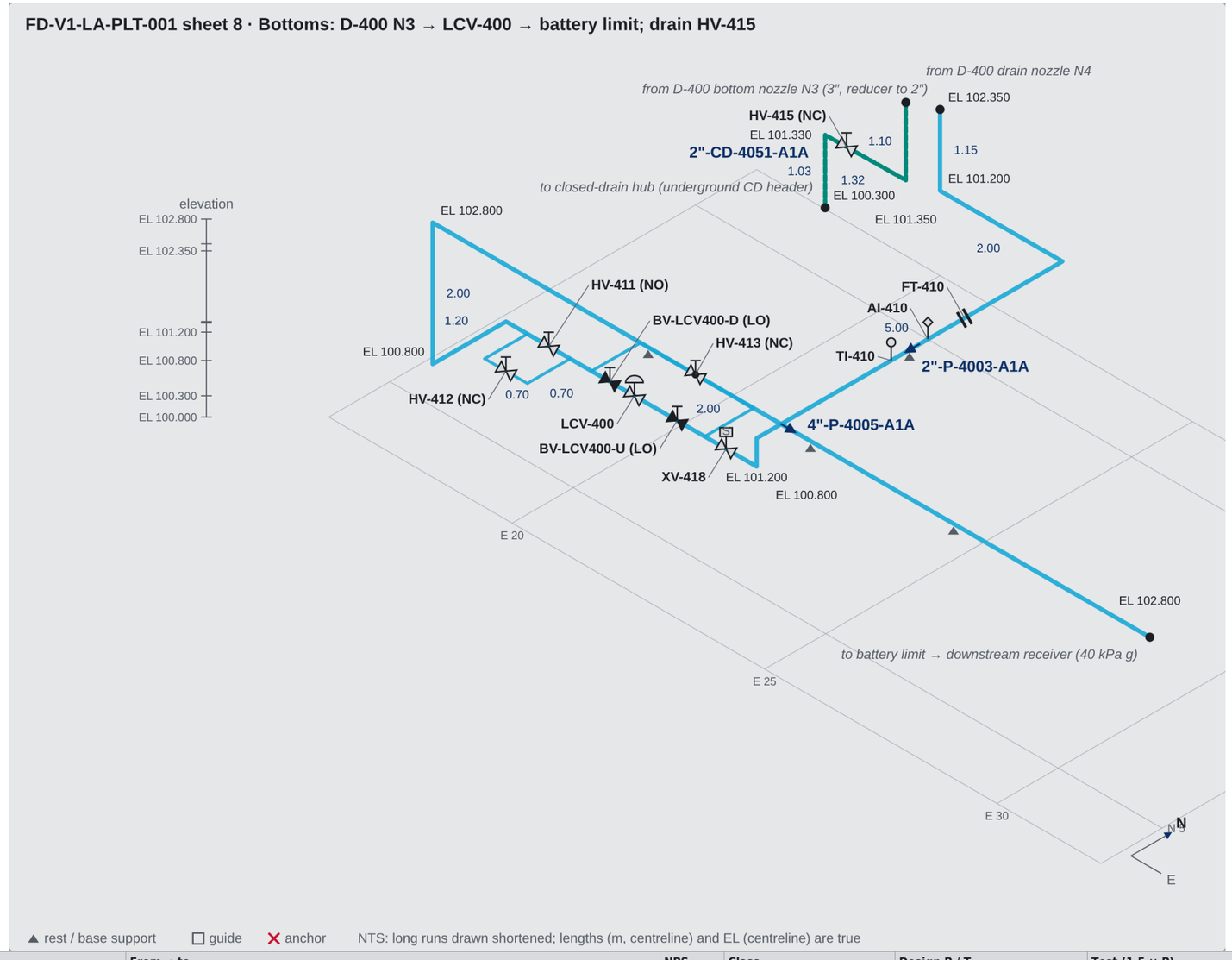
Bill of materials · 4"-PV-4021-A1A (b) · 4″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	4″	14.2	m	route length 13.5 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	4″	4	ea	
Tee, equal, BW	4″	2	ea	branch runs (bypass loops / branches)
Flange WN RF, class 150	4″	6	ea	equipment nozzles 0 + 2 per flanged valve
Gasket spiral-wound, class 150	4″	6	ea	
Stud bolt set, class 150	4″	6	set	
Gate valve, class 150 (station block, locked open)	4″	1	ea	BV-PCV420-D
Gate valve, class 150 (hand valve)	4″	2	ea	HV-421, HV-422
Support: Rest (shoe)	4″	1	ea	

Bill of materials · 3"-PV-4023-A1A · 3″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	3″	0.3	m	route length 0.2 m + 5 % (cut and weld allowance)
Flange WN RF, class 150	3″	2	ea	equipment nozzles 1 + 2 per flanged valve
Gasket spiral-wound, class 150	3″	2	ea	
Stud bolt set, class 150	3″	2	set	
Relief valve (by instruments)	3″	1	ea	PSV-400

Sheet 8 · Bottoms: D-400 N3 → LCV-400 → battery limit; drain HV-415



Line	Service	From → to	NPS	Class	Design P / T	Test (1.5 × P)	Insulation	Route length	Line list
2"-P-4003-A1A	Drum liquid at its bubble point	D-400 bottom nozzle N3 (3″, reducer to 2″) → LCV-400	2″	A1A · class 150	450 kPa g / 150 °C	675 kPa g	— (none)	14.0 m	✓
4"-P-4005-A1A	Bottoms after LCV-400, flashing (drawn as part of 2"-P-4003-A1A)	LCV-400 outlet → battery limit → downstream receiver (40 kPa g)	4″	A1A · class 150	450 kPa g / 150 °C	675 kPa g	— (none)	21.5 m	✓ proposed
2"-CD-4051-A1A	Drum drain, intermittent (not numbered on the P&ID)	D-400 drain nozzle N4 → closed-drain hub (underground CD header)	2″	A1A · class 150	450 kPa g / 150 °C	675 kPa g	— (none)	3.5 m	✓ proposed

Worked example (2"-P-4003-A1A). Route length = sum of the centreline segments of the main run: 1.15 + 2.00 + 5.00 + 0.40 + 2.00 = 10.55 m, plus 1 branch run(s) 3.45 m; ordered = route × 1.05 (cut and weld allowance, assumed) = 14.7 m. Supports: rests at the suggested span for 2″ (3 m, typical) on horizontal runs, guides every 6 m on risers, the rack bents on the rack: 1 supports. Hydrotest (ASME B31.3, 1.5 × design, temperature correction ignored) = 1.5 × 450 = 675 kPa g.

Bill of materials · 2"-P-4003-A1A · 2″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	2″	14.7	m	route length 14.0 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	2″	6	ea	
Tee, equal, BW	2″	2	ea	branch runs (bypass loops / branches)
Flange WN RF, class 150	2″	9	ea	equipment nozzles 1 + 2 per flanged valve
Orifice flange pair WN RF class 150 + plate	2″	1	set	FT-410
Gasket spiral-wound, class 150	2″	11	ea	
Stud bolt set, class 150	2″	11	set	
Gate valve, class 150 (station block, locked open)	2″	1	ea	BV-LCV400-U
Globe valve, class 150 (throttling bypass)	2″	1	ea	HV-413
Control valve, class 150 (by instruments)	2″	1	ea	LCV-400
Ball valve, class 150, spring-return actuator (by instruments)	2″	1	ea	XV-418

Item	Size	Qty	Unit	Note
Thermowell 1½″ flanged	1½″	1	ea	TI-410
Sample connection: ¾″ weldolet + root valve	¾″	1	ea	AI-410
Support: Rest (shoe)	2″	1	ea	

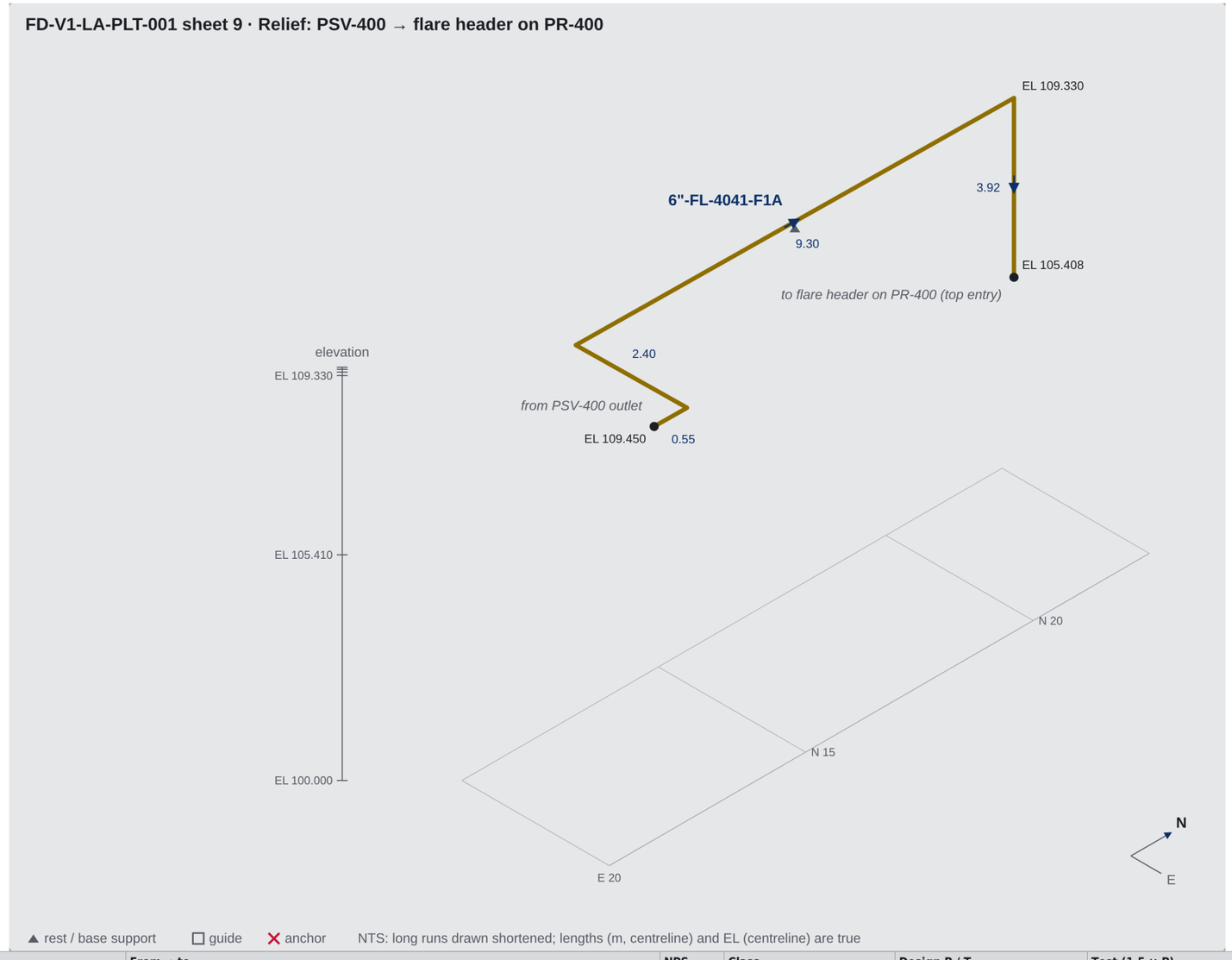
Bill of materials · 4"-P-4005-A1A · 4″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	4″	22.6	m	route length 21.5 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	4″	5	ea	
Tee, equal, BW	4″	2	ea	branch runs (bypass loops / branches)
Flange WN RF, class 150	4″	6	ea	equipment nozzles 0 + 2 per flanged valve
Gasket spiral-wound, class 150	4″	6	ea	
Stud bolt set, class 150	4″	6	set	
Gate valve, class 150 (station block, locked open)	4″	1	ea	BV-LCV400-D
Gate valve, class 150 (hand valve)	4″	2	ea	HV-411, HV-412
Support: Rest (shoe)	4″	3	ea	

Bill of materials · 2"-CD-4051-A1A · 2″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	2″	3.7	m	route length 3.5 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	2″	2	ea	
Flange WN RF, class 150	2″	3	ea	equipment nozzles 1 + 2 per flanged valve
Gasket spiral-wound, class 150	2″	3	ea	
Stud bolt set, class 150	2″	3	set	
Gate valve, class 150 (hand valve)	2″	1	ea	HV-415

Sheet 9 · Relief: PSV-400 → flare header on PR-400



Line	Service	From → to	NPS	Class	Design P / T	Test (1.5 × P)	Insulation	Route length	Line list
6"-FL-4041-F1A	PSV-400 tail pipe (not numbered on the P&ID)	PSV-400 outlet → flare header on PR-400 (top entry)	6″	F1A · class 150	350 kPa g / 150 °C	525 kPa g	— (none)	16.2 m	✓ proposed

Worked example (6"-FL-4041-F1A). Route length = sum of the centreline segments of the main run: 0.55 + 2.40 + 9.30 + 3.92 = 16.17 m; ordered = route × 1.05 (cut and weld allowance, assumed) = 17.0 m. Supports: rests at the suggested span for 6″ (5.2 m, typical) on horizontal runs, guides every 6 m on risers, the rack bents on the rack: 1 supports. Hydrotest (ASME B31.3, 1.5 × design, temperature correction ignored) = 1.5 × 350 = 525 kPa g.

Bill of materials · 6"-FL-4041-F1A · 6″ · ASTM A106 Gr. B seamless (assumed) · Sch 40 · ASME class 150

Item	Size	Qty	Unit	Note
Pipe, ASTM A106 Gr. B seamless (assumed), Sch 40	6″	17.0	m	route length 16.2 m + 5 % (cut and weld allowance)
Elbow 90° LR, BW, Sch 40	6″	3	ea	
Flange WN RF, class 150	6″	1	ea	equipment nozzles 1 + 2 per flanged valve
Gasket spiral-wound, class 150	6″	1	ea	
Stud bolt set, class 150	6″	1	set	
Support: Rest (shoe)	6″	1	ea	

Sheet 10 · Operator round, hand valves and layout.json

No.	Station	Floor	Camera p (x, y, z)	Look at	Items
1	Unit entrance from the east road (north aisle)	grade	31.00, 16.60, 0.00	20.0, 11.0, 4.0	D-400
2	Hot-oil station: XV-409, TCV-401, supply gauges	grade	9.50, 15.00, 0.00	9.5, 14.0, 1.0	XV-409, TI-402, PDI-406, PG-406, TCV-401, FT-405
3	Feed preheater E-400 and the feed thermometers	grade	7.00, 12.70, 0.00	9.2, 11.4, 1.4	E-400, TI-400, FT-400, TI-403, TT-401
4	Feed supply P-400 at the battery limit	grade	4.50, 8.80, 0.00	3.0, 6.3, 0.6	P-400
5	XV-408 and the FCV-400 let-down station	grade	13.90, 9.20, 0.00	14.6, 11.0, 1.0	XV-408, BV-FCV400-U, FCV-400, BV-FCV400-D, HV-403, HV-401, HV-402
6	D-400 skirt and the drain HV-415	grade	18.60, 9.60, 0.00	19.2, 10.7, 1.4	D-400, HV-415
7	D-400 east side: LG-400, LT-400, LSLL-417	grade	21.00, 10.15, 0.00	21.2, 11.0, 3.6	LG-400, LT-400, LSLL-417
8	Bottoms station: XV-418, LCV-400, HV-411/412/413	grade	20.00, 7.50, 0.00	20.3, 6.2, 0.8	FT-410, AI-410, TI-410, XV-418, BV-LCV400-U, LCV-400, BV-LCV400-D, HV-413, HV-411, HV-412
9	D-400 platform PL-1 (EL 105.600): N1, M1, LSHH-416	PL-1 EL 105.600	20.95, 9.75, 5.60	20.6, 11.6, 6.4	LSHH-416
10	D-400 top platform PL-2 (EL 108.000): PSV-400, PG-420	PL-2 EL 108.000	21.15, 10.25, 8.00	20.2, 11.6, 9.3	PSV-400, PG-420, PT-420, PSHH-426, AI-400
11	PCV-420 station on S-400 (EL 108.000)	S-400-1 EL 108.000	24.00, 12.60, 8.00	24.0, 14.0, 9.0	FI-420, AI-420, BV-PCV420-U, PCV-420, BV-PCV420-D, HV-423, HV-421, HV-422
12	Nitrogen blanket PCV-430 / HV-430	grade	24.20, 7.20, 0.00	23.6, 8.0, 1.0	HV-430, PCV-430

Hand valves (layout.json → hv)

Hand valve	Model key	x, y, z m	Pipe axis	Normal	Description
BV-FCV400-U	—	14.55, 11.00, 1.00	x	LO	control-valve station isolation valve, locked open (P&ID note 5); unnumbered on the P&ID (query LA-07)
HV-403	HV403	15.00, 11.70, 1.00	x	NC (throttling)	globe bypass (throttling), normally closed
BV-FCV400-D	—	15.40, 11.00, 1.00	x	LO	control-valve station isolation valve, locked open (P&ID note 5); unnumbered on the P&ID (query LA-07)
HV-401	HV401	16.25, 11.00, 1.00	x	NO	hand valve, normally open
HV-402	HV402	16.25, 10.40, 1.00	x	NC	hand valve, normally closed
BV-PCV420-U	—	23.60, 14.00, 9.00	x	LO	control-valve station isolation valve, locked open (P&ID note 5); unnumbered on the P&ID (query LA-07)
HV-423	HV423	24.00, 13.30, 9.00	x	NC (throttling)	globe bypass (throttling), normally closed
BV-PCV420-D	—	24.40, 14.00, 9.00	x	LO	control-valve station isolation valve, locked open (P&ID note 5); unnumbered on the P&ID (query LA-07)
HV-421	HV421	25.30, 14.60, 9.00	y	NO	hand valve, normally open
HV-422	HV422	24.80, 14.60, 9.00	y	NC	hand valve, normally closed
BV-LCV400-U	—	20.70, 6.00, 0.80	x	LO	control-valve station isolation valve, locked open (P&ID note 5); unnumbered on the P&ID (query LA-07)
HV-413	HV413	20.20, 6.80, 0.80	x	NC (throttling)	globe bypass (throttling), normally closed
BV-LCV400-D	—	19.60, 6.00, 0.80	x	LO	control-valve station isolation valve, locked open (P&ID note 5); unnumbered on the P&ID (query LA-07)
HV-411	HV411	18.60, 6.00, 0.80	x	NO	hand valve, normally open
HV-412	HV412	18.60, 5.30, 0.80	x	NC	hand valve, normally closed
HV-415	HV415	18.75, 10.72, 1.34	x	NC	hand valve, normally closed
HV-430	HV430	24.20, 8.00, 1.00	x	NO	hand valve, normally open

layout.json keys

Key	Count	Content
frame	—	units, axes, origin, grade EL 100.000, wind from 225° (assumed), angle conventions
equipment	5	tag, name, class, shape (vcyl / hcyl / box), x, y, z (bottom), D, L or h, w, d, axis; D-400 skirt and tangent lines; E-400 channel end, saddles, pull direction; buildings off-plot
structs	2	pipe rack PR-400 (tier, bents, columns) and the steel structure S-400 (deck, columns)
platforms	3	PL-1 and PL-2 on D-400 (ring: z, inner and outer radius, arc a0–a1) and the S-400 deck, each with what it serves
ladders / stairs	2 / 1	position, from and to elevation, outward angle, cage; stair footprint with landing
areas	10	battery limit, roads, escape routes, access way under the rack, crane hardstand, bundle-pull space
lines	23	every line of the line list plus six rack headers: id, number, segment, service (colour), NPS, OD, insulation, runs (polylines of [x, y, z]), inline components with their distance along the run
tags	47	every field instrument, valve and gauge: tag, device, kind, x, y, z of the head / hand wheel, the line point it sits on, the live reading key meas (sim.meas) and unit, and for hand valves the model key and normal position
hv	17	the hand valves in the format of V3_HV: tag, model key (sim.P.hv), position, pipe axis, normal position NO / NC, throttling flag (11 with a model key, the 11 of FIELDV, and 6 locked-open station blocks without one)
route	12	stations of the outside-operator round: number, name, floor, camera point p, look target, via waypoints from the previous station, items (tags and equipment)

Sheet 11 · Checks against the model and the other documents

Check	Layout value	Model / other document	Result	Note
Every item of the equipment list FD-V1-ME-EQL-001 is on the plot plan (equipment or valve tag)	12 / 12	D-400, E-400, P-400, FCV-400, TCV-401, LCV-400, PCV-420, PCV-430, PSV-400, XV-408, XV-409, XV-418	✓	
Every position of the shared plot plan (core/plot.json) is unchanged	12 / 12	plot.json	✓	
No two equipment bodies overlap; no body cuts another item's platform or deck	0 overlaps	3 bodies, 3 floors	✓	
Every line number on an isometric exists in the line list FD-V1-PE-LIN-001	10 / 10	17 lines in the line list	✓	
Isometric size and design conditions = line list	10 / 10	NPS, design P, design T	✓	
Every line of the line list is routed in layout.json	17 / 17	line list	✓	
Every inline component sits on its pipe run (≤ 2 cm)	0 off	all lines	✓	
Every tag marked "on the P&ID" is in the P&ID / instrument list; the others are listed for numbering	39 on P&ID, 8 to number	FD-400-PID-001, instrument list	✓	to number: BV-FCV400-U, BV-FCV400-D, PG-420, BV-PCV420-U, BV-PCV420-D, BV-LCV400-U, BV-LCV400-D, PG-406 (LA-07)
Every hand valve of FIELDV has a 3-D position in layout.json	11 / 11	FIELDV: HV401, HV402, HV403, HV411, HV412, HV413, HV421, HV422, HV423, HV430, HV415	✓	
Every SIS valve of the model (sim.P.xv) has a 3-D position	3 / 3	XV-408, XV-409, XV-418	✓	
Normal positions of the hand valves = the design snapshot (snapDesign.P.hv)	11 / 11	NO = open at design	✓	
Every route station is reachable: on a floor, legs clear, level changes on ladders / stair	12 / 12 stations	51 items on the round	✓	
Every route item exists in the layout (tag or equipment)	0 missing	route	✓	
Spacing and layout rules of part 3 satisfied	21 / 25	typical practice (assumed)	✗	SR-06, SR-07, SR-08, SR-09: plot.json positions → LA-05
Every field reading point and hand wheel within reach of grade or a floor (2.0 m; LG 6 m)	37 / 38	LSLL-417 3.45 m	✗	LA-04
Model static height K.Hel = layout height of the drum bottom above LCV-400	3.00 m	1.80 m (EL 102.600 − 100.800)	✗	LA-02 (answers EM-01)

Check	Layout value	Model / other document	Result	Note
Bottoms liquid stays above its bubble pressure at FT-410: design flow, NLL, 2″	9.4 kPa margin	> 0	✓	
… at the FIC-410 high alarm (18 m³/h) and LAL (20 %), 2″ as drawn	−3.6 kPa margin	> 0 (4″: 2.7 kPa)	✗	LA-06, EP-03
PSV-400 tail-pipe equivalent length from the layout ≤ the relief-basis assumption	25.4 m (16.2 m + 3 elbows × 20 D)	45 m (FD-V1-PE-RLB-001, ER-07)	✓	
PSV-400 inlet branch from the layout ≤ the relief-basis route (3″ branch 0.8 m)	0.20 m	0.8 m (inlet-loss basis)	✓	
2″ feed route (P-4001 + P-4002 to FCV-400) within the model's lumped line loss (≈ 19 m of 2″, EP-10)	51.6 m equivalent (as plot.json)	32.4 m with XV-408 per LA-01	✗	EP-10 (3″), LA-01
layout.json validates (same checks as tools/test.js)	valid	placed, overlaps, route, tags, lines, inline items	✓	

17 ✓, 5 ✗. Each ✗ has a query in part 8.

Engineering queries raised by layout

No.	Document	Subject	Query	Why it matters	Proposed answer	Status
LA-01	core/plot.json	XV-408 position on the plot plan	plot.json puts XV-408 at (13, 11), east of E-400 and 2 m west of FCV-400, i.e. on the hot feed between E-400 and FCV-400. FD-400-PID-001 sheet 1 draws XV-408 FC → FT-400 → TI-400 → E-400 → TT-401 → FCV-400 → HV-401 → D-400; the line list carries XV-408 on 2"-P-4001-A1A (cold feed) and the equipment list rates it for 65 °C. To keep plot.json, the cold feed has to run past E-400 to XV-408 and back to the west channel: 2"-P-4001-A1A is routed 25.8 m long.	HAZOP R-02 (thermal relief between XV-408 and FCV-400) and the Consequence-analysis S1 trapped volume both assume the P&ID order; the 3-D model and the field rounds would show XV-408 on the wrong line.	Move XV-408 to (4.6, 7.6), EL 101.000, on 2"-P-4001-A1A near the battery limit, with FT-400 downstream (dashed on the plot plan): the cold feed becomes 10.8 m. Update plot.json and re-run the Consequence-analysis distances for S1 (the release point does not change).	OPEN
LA-02	model K.Hel, FD-V1-ME-DS-001 (EM-01)	Static head datum of the model (K.Hel)	In the model K.Hel = 3 m is used only in the LCV-400 pressure balance (head = ρg(K.Hel + h_liquid), h_liquid measured from the drum bottom), so it is the height of the drum bottom above the LCV-400 inlet, not above grade. With the bottom tangent at EL 103.000 (plot.json) the drum bottom is at EL 102.600 and LCV-400 at EL 100.800: 1.80 m.	EM-01 compares K.Hel with the bottom tangent and the drum bottom above grade; neither is the datum the model uses. The difference changes the LCV-400 ΔP at design (part 4).	Answer to EM-01: keep the plot plan (bottom tangent EL 103.000); state in the model notes that K.Hel is the drum-bottom height above LCV-400 and set it to the layout value in the next model revision (the LCV-400 opening at design moves by about 1 %, part 4). Until then the model is 1.2 m optimistic on the bottoms head.	OPEN
LA-03	core/plot.json, FD-V1-PE-LIN-001 (EP-01)	Two-phase riser after FCV-400	plot.json puts FCV-400 at grade (EL 101.000); the let-down outlet 6"-P-4004-A1A then rises 5.9 m to N1 (EL 106.900) carrying a flashing two-phase mixture (17 wt% vapour at design).	Typical practice places a let-down valve close to the drum and at about the inlet-nozzle elevation so that the flashing section is short and does not rise (slug flow, vibration, erosion at the elbows).	Keep FCV-400 at grade for operability (FIC-400 and the HV-401/402/403 bypasses are worked from grade) only after a two-phase flow-regime and vibration check of the 6″ riser; otherwise move the FCV-400 station to PL-1 (EL 105.600) with a 2″ hot-liquid riser below it.	OPEN
LA-04	FD-400-PID-001, SRS	LSLL-417 not reachable for proof testing	LSLL-417 sits on N8 at EL 103.450, 3.45 m above grade and below PL-1; the bridle bottom N6B is at EL 103.150. Neither is within 2.0 m of a floor.	The SIS needs LSLL-417 (I-2) proof-tested at the interval of the SRS; work at height without a platform needs a mobile elevating work platform each time.	Add a small access platform on the SE quadrant of the skirt at about EL 102.000 (reached by LD-1), or fit LSLL-417 as a DP / guided-wave transmitter with its electronics at grade.	OPEN
LA-05	core/plot.json	Siting of the control building and the substation	Distances to the nearest hydrocarbon item: control building and substation (part 3, SR-06 to SR-09). The control building lies east, across the prevailing south-west wind, and the substation north-east, downwind of the unit.	Typical spacing tables ask 30–60 m from light-hydrocarbon process equipment to a non-blast-resistant occupied building and about 30 m to ignition sources, with both upwind. The Consequence-analysis tab finds the control building inside the LFL cloud of S1, S2 and S5.	Building siting study (blast, gas ingress, fire) before the layout is frozen; move the substation upwind (south-west) or provide gas detection at its air intakes; the control building as in the Consequence-analysis findings.	OPEN
LA-06	FD-V1-PE-LIN-001 (EP-03)	Bottoms line flashes at FT-410 at high flow and low level	With 2"-P-4003-A1A the static head between the drum liquid and FT-410 does not cover the entrance, friction and orifice losses at the FIC-410 high-alarm flow and a low drum level: the vena contracta of FT-410 drops below the bubble pressure (part 4).	Flashing in the orifice gives a false (low, noisy) flow reading to FIC-410 / LIC-400 (Rev B cascade) and cavitation damage.	Make the bottoms line 4″ from N3 to LCV-400 (EP-03), keep FT-410 at the lowest point of the run, size the orifice for a low ΔP (≤ 10 kPa at full scale) or use a magnetic / Coriolis meter.	OPEN
LA-07	FD-400-PID-001	Field items without a tag number	The control-valve station block valves (P&ID note 5: locked open) and the local gauges the Field tab reads (PG-420 on the drum top, the hot-oil supply gauge) have no tag on the P&ID; the layout calls them BV-FCV400-U/D, BV-LCV400-U/D, BV-PCV420-U/D, PG-420 and PG-406.	Field rounds, isolation lists and the 3-D model need a unique tag for every valve and gauge the operator touches or reads.	Number them on the next P&ID revision (for example HV-404…HV-409 for the station blocks, PI-420 / PI-406 for the gauges) and update layout.json.	OPEN